Question

Adding sodium hydroxide to an aqueous solution of a nitrate $\mathbf{M}(\mathrm{NO_3})_\mathrm{x}$ of a certain metal $\mathbf{M}$ yields a precipitate $\mathbf{X}$. Treating $\mathbf{X}$ and $\mathbf{M}(\mathrm{NO_3})_\mathrm{x}$ with a persulfate yields a precipitate $\mathbf{Y}$.

Hydrothermal synthesis of $\mathbf{Y}$ with M element under high pressure yields crystal $\mathbf{Z}$.

It is known that $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$ are all oxides of the metal $\mathbf{M}$, and the mass fraction of oxygen element in substance $\mathbf{X}$ is 6.9%, $\mathbf{Y}$ is 12.9%, and $\mathbf{Z}$ is 4.7%.

The crystal of $\mathbf{Z}$ can be approximately regarded as the stacking of different atomic layers in the manner of $\mathbf{......AcB\square AcB\square.......}$ Where $\mathbf{A}$ and $\mathbf{B}$ are close-packed layers of $\mathbf{M}$ atoms, $\mathbf{c}$ is an octahedral interstitial layer filled with some O atoms, and $\square$ is an empty octahedral interstitial layer.

The coordinates of all O atoms in the unit cell are:

$(\frac{2}{3}, \frac{1}{3}, \frac{1}{2})$

$(\frac{1}{3}, \frac{2}{3}, \frac{1}{2})$

And $\frac{2}{3}$ of the $\mathbf{M}$ atoms in the unit cell are located on the unit cell face

Which of the following statements are correct:

1.The occupancy rate of O atoms in the $\mathbf{c}$ layer is 1

2.The occupancy rate of O atoms in the $\mathbf{c}$ layer is $\frac{2}{3}$

3.The occupancy rate of O atoms in the $\mathbf{c}$ layer is $\frac{1}{3}$

4.The x, y coordinates of one of the $\mathbf{M}$ atoms are $(\frac{2}{3}, \frac{2}{3})$

5.The x, y coordinates of one of the $\mathbf{M}$ atoms are $(\frac{1}{2}, \frac{1}{2})$

6.The x, y coordinates of one of the $\mathbf{M}$ atoms are $(\frac{1}{3}, \frac{2}{3})$

A. 1,4

B. 1,5

C. 1,6

D. 2,4

E. 2,5

F. 2,6

G. 3,4

H. 3,5

I. 3,6

J. All of the above options are incorrect or the answer is incomplete.

Answer

Correct Answer: D

Detailed Explanation

$\mathbf{X}$ obtained by precipitating and drying with sodium hydroxide is one of the most common oxides of metal M, which can be expressed as Double subscripts: use braces to clarify, and its mass fraction of oxygen element $\omega_{\mathrm{O}}=\frac{q\times16}{p\,M+q\times16}=6.9\%$.

Assume the chemical formula of $\mathbf{X}$ to be MO, $\mathrm{M_2O_3}$, $\mathrm{MO_2}$, etc., and substitute them into the equation. Finally, it is found that when it is $\mathrm{M_2O}$, the molar mass of M is 107.9, which is consistent with Ag. Therefore, the metal M is Ag, and $\mathbf{X}$ is $\mathrm{Ag_2O}$. Substituting the mass fraction, it can be further deduced that $\mathbf{Y}$ is AgO, and $\mathbf{Z}$ is $\mathrm{Ag_3O}$.

Substitute into the question conditions for verification, which is consistent with the chemical properties of Ag.

CHECKPOINT 3 PTS

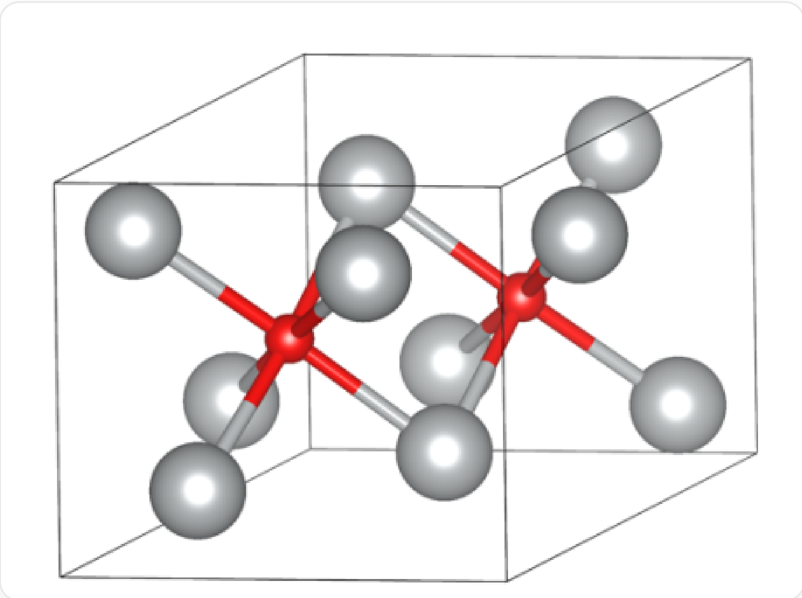
$\mathbf{X}$ is $\mathrm{Ag_2O}$, $\mathbf{Y}$ is AgO , $\mathbf{Z}$ is $\mathrm{Ag_3O}$

There are 2 oxygen atoms in the unit cell, so according to the chemical formula, there are 6 silver atoms, of which layers $\mathbf{A}$ and $\mathbf{B}$ have three silver atoms respectively. According to the ratio of octahedral voids to the number of atoms in the closest packing being 1:1, 6 silver atoms have a total of 6 octahedral voids. Since the oxygen atom interstitial layer and the void layer are spaced apart, there are 3 octahedral voids in layer $\mathbf{c}$, which are filled with 2 oxygen atoms, so the filling rate is $\frac{2}{3}$.

CHECKPOINT 2 PTS

The filling rate of O atoms in layer $\mathbf{c}$ is $\frac{2}{3}$, option 2 is correct, options 1 and 3 are incorrect

According to $\frac{2}{3}$ of the M atoms in the unit cell are located on the surface of the unit cell, there are 2 atoms on the surface of the unit cell and 1 atom inside the unit cell in each silver atom close-packed layer. According to the symmetry of the octahedron and the C_3 axis in the z direction through the oxygen atom, it can be deduced that the x and y coordinates of the silver atoms in layer $\mathbf{A}$ are $(\frac{1}{3}, 0)$, $(0, \frac{1}{3})$, $(\frac{2}{3}, \frac{2}{3})$, and the x and y coordinates of the silver atoms in layer $\mathbf{B}$ are $(\frac{2}{3}, 0)$, $(0, \frac{2}{3})$, $(\frac{2}{3}, \frac{2}{3})$ (layers $\mathbf{A}$ and $\mathbf{B}$ are interchangeable). The crystal diagram is as follows



Z的晶胞参考图：$\mathbf{A}$ 层银原子的x、y坐标为$(\frac{1}{3}, 0)$、$(0, \frac{1}{3})$、$(\frac{2}{3}, \frac{2}{3})$；$\mathbf{c}$ 层氧原子的坐标为 $(\frac{2}{3}, \frac{1}{3}, \frac{1}{2})$、$(\frac{1}{3}, \frac{2}{3}, \frac{1}{2})$；$\mathbf{B}$ 层银原子的x、y坐标为$(\frac{2}{3}, 0)$、$(0, \frac{2}{3})$、$(\frac{2}{3}, \frac{2}{3})$

CHECKPOINT 3 PTS

The possible x and y coordinates of silver atoms are $(\frac{1}{3},0)$, $(0,\frac{1}{3})$, $(\frac{2}{3},\frac{2}{3})$, $(\frac{2}{3},0)$, $(0,\frac{2}{3})$, $(\frac{2}{3},\frac{2}{3})$, so option 4 is correct, options 5 and 6 are incorrect

In summary, the answer is D

CHECKPOINT 1 PTS

The answer is D